MS1411: Mathematical Statistics

Week 5

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Lecture Objectives

- Understand the concept of the **Central Limit Theorem** (CLT)
- Learn how to construct **interval estimates** (confidence intervals) for a population mean, with both known and unknown variance
- Compare **one-sided** vs. **two-sided** confidence intervals
- Explore the difference between **Type I and Type II errors** in hypothesis testing
- Conduct basic **significance tests** for means, including **right-, left-, and two-tailed** tests
- Learn how to compute and interpret **p-values** (under normal or t-distributions)
- Gain familiarity with the Student's **t-distribution** for inference with unknown variance

Sampling Distributions and the Central Limit Theorem

2.1 Sampling Distribution of the Mean

- **Random Sampling:** Selecting n observations from a population to estimate characteristics (parameters) of that population.
- **Sample Mean $\bar{X}$:**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- **Sampling Distribution:** The probability distribution of $\bar{X}$ across many repeated samples of size n.

Sampling Distributions and the Central Limit Theorem

2.2 Central Limit Theorem (CLT)

- **Statement:** If $X1, X2, \ldots, Xn$ is a random sample from any population with mean μ and variance σ^2 , then, for large n, the distribution of

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

is approximately Standard Normal, $N(0,1)$, regardless of the population's original distribution (provided certain mild conditions hold).

- **Implications:**
    - For $n \geq 30$ (a common rule of thumb), $\bar{\boldsymbol{X}}$ is "approximately normal."
    - Forms the foundation for many confidence interval and hypothesis testing procedures.

Example (Intuitive): If you measure the average lifetime of light bulbs in a factory, each individual measurement might be non-normal (e.g., slightly skewed), but the average of *n* bulbs tends to be nearly normal for large *n*.

Example #1

- **Example #1: Central Limit Theorem**
- **Scenario:** A die is tossed 30 times, and this experiment is repeated 56 times to obtain 56 $\overline{X}$ values.
- **Expected Results:**
  - **Expected Mean:** 3.5
  - **Variance:** 2.91
  - **Results for 56 repetitions:** $\overline{X} = 3.5370$, $\sigma^2 = 2.91$
  - **Results for 100 repetitions:** $\overline{X} = 3.5068$, $\sigma^2 = 2.91$

Interval Estimation

3.1 Confidence Intervals: General Concept

- A **confidence interval (CI)** provides a range $[L1,L2]$ in which we expect the true population parameter (e.g., μ) to lie with high probability, say (1−α)×100%.
- Common confidence levels: **90%**, **95%**, **99%**.

3.2 Interval for a Population Mean: Variance Known

- **Assumptions**:
    - Population variance σ^2 is known or sample size n is large.
    - Population or sample size large enough so that CLT applies.
- **Z-Interval**:

$$\bar{x} \pm z_{\frac{\alpha}{2}}\left(\frac{\sigma}{\sqrt{n}}\right),$$

Where $z_{\frac{\alpha}{2}}$ is the critical value from the standard normal distribution for confidence level (1- α).

Interval Estimation

3.3 Interval for a Population Mean: Variance Unknown

- **When σ^2 is unknown**, replace σ with the **sample standard deviation** s.
- Use **Student's t-Distribution** with n−1 degrees of freedom:

$$\bar{x} \pm t_{\frac{\alpha}{2},n-1} \frac{s}{\sqrt{n}}$$

- Valid if population is **normally distributed** or n is reasonably large.

3.4 One- and Two-Sided Confidence Bounds

- **Two-Sided**: $[\bar{x} - E, \bar{x} + E]$
- **Upper One-Sided Bound**: $\bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}$
- **Lower One-Sided Bound**: $\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}$
- Choice depends on the **problem context** (e.g., a known minimum vs. unknown maximum threshold).

3.5 Inference with Unknown Mean and Unknown Variance

- For small samples from a **normal population**, use **t-interval** with $n-1$ degrees of freedom.
- For large samples (e.g., $n \geq 30$), the normal approximation (z-interval) is often adequate by CLT.

Practical Point: Always check if n is small and data are from a normal distribution, or if n is large enough to rely on CLT.

Example #2: Probability Calculation

- **Scenario:** A factory produces LED lamps with a life-span normally distributed with mean 800 hours and standard deviation 60 hours.
- **Task:** Find the probability that a random sample of 30 lamps will have an average life-span of less than 650 hours.

Solution:

- **Mean (μ):** 800 hours
- **Standard Deviation ($\sigma/\sqrt{n}$):** $60/\sqrt{30} = 10.95$
- **Z-Score Calculation:** $Z = (650 - 800) / 10.95 = -13.7$
- **Probability:** $P[\overline{X} < 650] = P[Z < -13.7] < 0.0002$

Example #3: Sample Size Calculation

- **Scenario:** Determine the sample size required to be 95% confident that the estimate of μ with population standard deviation $\sigma = 0.3$ is off by less than 0.05.

Solution:

- **Formula:** $n = (Z\alpha/2 * \sigma / e)^2$
- **Calculation:** $n = (1.96 * 0.3 / 0.05)^2 = 138.3 \approx 139$

Example #4: Confidence Interval Calculation

- **Scenario:** A new treatment is being investigated to prolong the average survival time of patients diagnosed with cancer. The survival time is normally distributed with a standard deviation of 3 months and an estimated expected survival time of 13 months.

Solution:

- **Confidence Interval Calculation:**
    - **95% Confidence Interval:** $P[1.96 \le Z \le 1.96] = 0.95$
    - **Interval:** Assuming $\overline{X} = 13$, $\sigma = 3$, and $n = 16$:
        - $L = \overline{X} - 1.96 * (\sigma/\sqrt{n}) = 13 - 1.47$
        - $U = \overline{X} + 1.96 * (\sigma/\sqrt{n}) = 13 + 1.47$
        - **Interval:** [11.53, 14.47]

Example #5: One-Sided Confidence Bound

- **Scenario:** In a psychological testing experiment, 25 subjects are selected randomly, and their reaction time to a stimulus is measured. The variance in reaction times is 4, and the average time is 6.2 seconds.

Solution:

- **Upper 95% Bound Calculation:**
  - **Formula:** $\bar{X} + Z\alpha * (\sigma/\sqrt{n})$
  - **Calculation:** $6.2 + (1.645 * \sqrt{4}/\sqrt{25}) = 6.858$ seconds

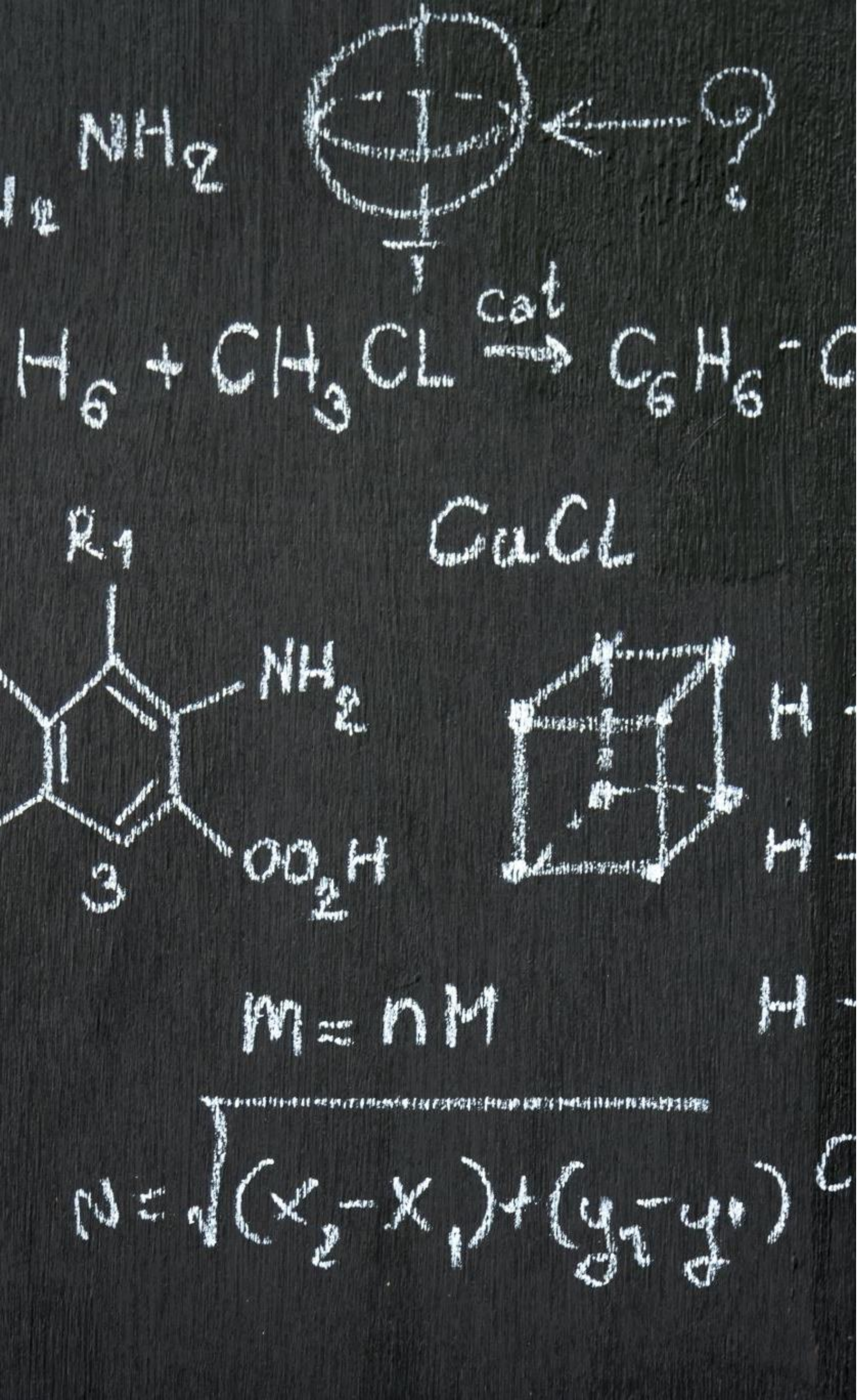
Interval Estimation

Inferences on the Mean and Variance of a Distribution

- Overview: Inferences on the Mean and Variance
  - Common Problem: Inference of mean and variance.
  - Confidence Interval: Provides an interval of values for a given parameter, $\theta \in [L_1, L_2]$.
- Estimation: Interval Estimation of Variance
  - Theorem: Let $X_1, X_2, \ldots, X_n$ be a random sample from a normal distribution with mean μ and variance σ^2. The limits for a $100(1 - \alpha)\%$ interval on σ^2 are given by:
    - $(n - 1)S^2 / b \leq \sigma^2 \leq (n - 1)S^2 / a$

Example #6: Confidence Interval for Standard Deviation

- Scenario: A package of coffee should weigh 500 g. A quality test randomly samples 20 packs off the line. The sampled variance is 10.
- Solution:
- Chi-Square Table Values:
- $a = \chi^2(0.975, 19) = 8.91$
- $b = \chi^2(0.025, 19) = 32.85$
- Interval Calculation:
- $(19 * 10 / 32.85) \leq \sigma^2 \leq (19 * 10 / 8.91)$
- Standard Deviation Interval: $2.40 \leq \sigma \leq 4.62$ grams



Interval Estimation

Estimation: Interval Estimation of Mean

- Known Variance: $Z = (\bar{X} - \mu) / (\sigma/\sqrt{n})$
- Unknown Variance: Substitute σ with its point estimator (S):
  - $\bar{X} \pm t\alpha/2 * (S/\sqrt{n})$

T-Distribution:

- Definition: The random variable T is defined as the ratio between a standard normal random variable and the square root of an independent chi-square random variable divided by the number of degrees of freedom.
- Formula: $T = Z / \sqrt{(\chi^2/v)}$

Example #7: Confidence Interval for Mean

Scenario: A random sample of 16 Americans yielded the following data on the number of pounds of beef consumed per year: 118, 115, 125, 110, 112, 130, 117, 112, 115, 120, 113, 118, 119, 122, 123, 126.

Solution:

T-Distribution Table: $t_{0.025,15} = 2.1314$

Calculation: $\bar{X} \pm t_{\alpha/2} * (S/\sqrt{n})$

- $\bar{X} = 118.44$
- $S = 5.66$
- Interval: $115.42 \leq \mu \leq 121.46$

Introduction to Hypothesis Testing

4.1 Basic Terminology

- **Statistical Hypothesis:** A claim or statement about a population parameter.
- **Null Hypothesis ($H0$):** Baseline claim, typically representing "no effect" or "status quo."
- **Alternative Hypothesis ($H1$or Ha):** Statement we suspect may be true if we reject $H0$.
  - Example:
    - $H0$: μ=10 (the process mean is 10)
    - $H1$: μ≠10

4.2 Type I and Type II Errors

- **Type I Error (α):** Rejecting $H0$ when $H0$ is actually true.
- **Type II Error (β):** Failing to reject $H0$ when $H0$ is false (thus $H1$ is true, but we did not detect it).
- **Common practice:** Fix α at a small level (e.g., 0.05). Minimizing β is also important but often more complex in multi-parameter settings.

Introduction to Hypothesis Testing

4.3 Significance Testing and p-Values

- **Significance Level (α):** The probability of a Type I error the analyst is willing to tolerate.
  - **Standard Values for α:** 0.01, 0.05, 0.1
- **Test Statistic:** A function of sample data used to decide whether to reject $H0$.
- **p-Value:** The probability (under $H0$) of observing a test statistic as extreme or more extreme than the actual sample.
  - If p-value < α, reject $H0$.
  - A small p-value indicates the data strongly contradict $H0$.

Example: Testing μ=100 vs. μ≠100. Suppose:

$$Z = \frac{\bar{X} - 100}{\frac{\sigma}{\sqrt{n}}} = 2.4$$

and the two-sided p-value might be 0.016. If α=0.05, we reject $H0$.

Introduction to Hypothesis Testing

4.4 Right-, Left-, and Two-Tailed Tests

- Right-tailed test: $H1$:μ>μ0. Region of rejection is large values of $\bar{X}$.
- Left-tailed test: $H1$:μ<μ0. Region of rejection is small values of $\bar{X}$.
- Two-tailed test: $H1$:μ≠μ0. Region of rejection is $\bar{X}$ sufficiently far above or below μ0.

4.5 Connection to z-Distribution and t-Distribution

- z-test for large n or known σ:

$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

- t-test for small n when σ unknown and data from normal population:

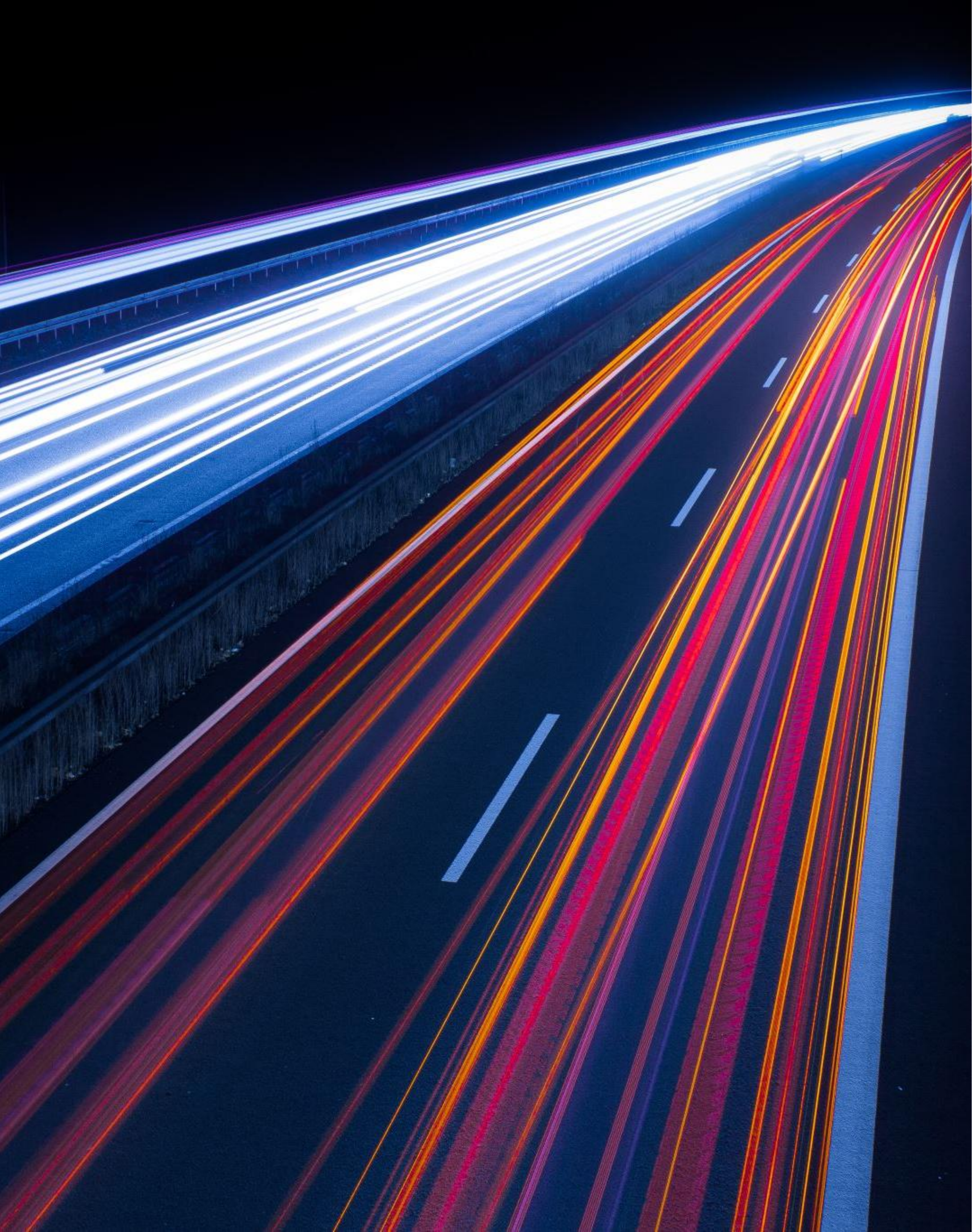
$$T = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Example #8

Hypothesis Testing

- Scenario: A chemical engineer claims that the population mean of a certain batch process is 500 grams per millimeter of raw material. To check this claim, she samples 25 batches each month.

- Solution:
    - T-Value Calculation: $t_{0.05,24} = 1.711$
    - Interval Calculation: $\bar{X} \pm t_{\alpha/2} * (S/\sqrt{n})$
    - $\bar{X} = 518$
    - $S = 40$
    - Interval: $495.6 \leq \mu \leq 540.4$
    - Conclusion: 500 is part of the interval, so the process should not be adjusted.



Example #9: Hypothesis Testing

- **Scenario:** Highway engineers found that more than 50% of vehicles have misaimed headlights. If supported, a new inspection program will be launched.
- **Solution:**
  - **Hypotheses:**
    - H_0: $p = 0.5$
    - H_1: $p > 0.5$
  - **Errors:**
    - **Type I Error:** Launch inspection program without need (false positive).
    - **Type II Error:** Fail to launch inspection program when needed (false negative).

Class Activity

1. **Objective**:
    1. Compute the sample mean $\bar{x}$ and sample standard deviation s.
    2. Construct a 95% confidence interval for the population mean, μ.
    3. Suppose a historical or "advertised" mean $\mu 0$ is given; test $H0$:μ=$\mu 0$ vs. $H1$:μ≠$\mu 0$.
    4. Compute and interpret the p-value; decide whether to reject $H0$.
2. **Data Collection**:
    1. Assume we have a process whose "advertised" or historical mean is $\mu 0$=50. We take a sample of n=9 items (measurements, reaction times, lengths, or any continuous variable). Our sample data (in arbitrary units) might be:

    49, 51, 48, 49, 52, 54, 47, 50, 51.

Solution Class Activity

- 3. Analysis
  - 3.1 Compute $\bar{x}$ and s
    - **Sample Mean**,
      - $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$
      - Add them all up: 49+51+48+49+52+54+47+50+51 = 451.
      - Then divide by n=9.
      - $\bar{x} = \frac{451}{9} = 50.111$
    - **Sample Standard Deviation**, s, is found from
      - $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, s = \sqrt{s^2}$
      - Compute each $(x_i - \bar{x})^2$, sum them, then divide by $n - 1$
      - In this example, $s^2 = 4.61$ and $s = 2.147$
  - 3.2 Construct a 95% Confidence Interval for μ
    - Because n is small (9 observations), we assume the population is approximately normal and use a t-distribution with n−1=8 degrees of freedom.
    - **The t-critical value** for a 95% two-sided confidence interval and 8 d.f. is roughly $t_{0.025,8}$≈2.306.
    - The standard error of the mean is
      - $SE = \frac{s}{\sqrt{n}} = \frac{2.147}{3} = 0.716$
    - The **margin of error** at 95% confidence is
      - $E = t_{0.025,8} \times SE = 2.306 \times 0.716 = 1.652$
    - **Confidence Interval**:
      - $\bar{x} \pm E = 50.111 \pm 1.652 = (48.459, 51.763)$
      - **Interpretation**: We are 95% confident that the true mean μ lies between approximately **48.46** and **51.76**.

Solution Class Activity

- 3.3 Test the Hypothesis $H0$:μ=50 vs. $H1$:$\mu \neq 50$
  - **Significance Level**: We choose α=0.05 (common in many settings).
  - **Test Statistic**: Because σ is unknown and *n* is small, we use the **t-statistic**:
    - $T = \frac{\bar{X}-\mu_0}{\frac{s}{\sqrt{n}}} = \frac{(50.111-50)}{\frac{2.147}{\sqrt{9}}} = \frac{0.111}{\frac{2.147}{3}} = 0.155$
  - Under $H0$, T follows a t-distribution with 8 d.f.
- 3.3.1 p-Value
  - For a two-sided test, the p-value is $P(|T| \geq 0.155)$ with df = 8.
  - A small T-value (~0.155) is very close to zero, meaning the sample mean is not far from 50 at all.
  - Checking t-tables or software, we would see this p-value is quite large (above 0.80).
- 3.3.2 Decision
  - If p-value ≥α, we fail to reject $H0$.
  - Since the p-value is certainly above 0.05, we do not reject $H0$.
  - Conclusion: There is no evidence that $\mu \neq 50$μ=50.

Solution Class Activity

- **4. Discussion**
  - The result is consistent with the 95% confidence interval. Notice that 50 (the hypothesized mean) lies squarely inside the confidence interval (48.46, 51.76).
  - Because the confidence interval includes 50, it's no surprise that the p-value is large and we do not reject $H0$.
  - Thus, using these 9 measurements, we have no reason to doubt the manufacturer's claim of a 50-unit mean.

Take-Home Message:

- Confidence intervals and hypothesis tests are closely related: if $\mu 0$ lies inside the 95% CI, then a two-sided test at α=0.05 will not reject $H0$.
- Checking assumptions:
  - We assumed the data come from a normal population (small sample).
  - For a large sample, you could use the z-based procedure (because of the Central Limit Theorem).

Summary

Inferences on the Mean and Variance of a Distribution:

- Sampling Distribution of X
- Central Limit Theorem
- Interval Estimation

Interval Estimation:

- Overview
- Interval Estimation of Variance
- Interval Estimation of Mean

Hypothesis Testing:

- Significance Testing
- Types of Errors (Type I and Type II)
- P-Value and Power of the Test

Examples:

- Various practical examples illustrating concepts like Central Limit Theorem, Confidence Intervals, and Hypothesis Testing.